Definitions

Def: Constant map Let $C : R \to R[X]$ be the *constant homomorphism* which maps constants in R to polynomials with coefficients in R. Given by

$$a \in R \overset{C}{\mapsto} aX^0 \in R[X]$$

Link: Link here

The substitution map in lean4

Let $\phi : R \to R'$ be a ring homomorphism.

We can define a unique homomorphism $T : R[X] \to R'$ which agrees with ϕ on constant polynomials (polynomials of the form X^0) and maps all others $x \mapsto \alpha \in R'$. (This *substitution map* definition is borrowed from: Artin's Algebra Book)

Lean defines exactly such a map via the eval_2 definition:

```
irreducible_def eval2 (p : R[X]) : S :=
  p.sum fun e a => f a * x ^ e
```

Where sum is defined by

```
p.sum f = $\sum n \in$ p.support, f n (p.coeff n)
```

We will therefore notate the eval_2 function with the following

$$T_{\phi,\alpha} = \sum_{n \in \sup(p)} \phi(\cdot) \times (\cdot)^n : \quad p \in R[X] \quad \mapsto \quad \sum_{n \in \sup(p)} \phi(p_i)\alpha^n$$

Link: Eval_2 map

Observe that T is semantically equivalent with a *substitution* homomorphic map

Realizations:

Evaluation map Evaluating a polynomial at a particular value in lean4 is given by eval:

```
def eval (x : R) (p : R[X]) : R :=
  eval2 (RingHom.id _) x p
```

Let $\alpha \in R$, $p(x) \in R[X]$ and let I be the identity ring homomorphism. Then we can notationally represent the eval map as

$$T_{(I,\alpha)}(p(x)) = T_{I,\alpha}\left(\sum_k a_k x^k\right)$$

Prop: $T_{I,\alpha}(p)(x) = p(\alpha)$. That is, $T_{I,\alpha}$ gives the polynomial p as a function evaluated at value $\alpha \in R$

Proof: Observe, using the fact that T itself is homomorphic

$$T_{(I,\alpha)}(p(x)) = \sum_k T_{I,\alpha}\left(a_k x^k\right)$$

$$T_{(I,\alpha)}(p(x)) = \sum_k T_{I,\alpha}\left(a_k\right) \cdot T_{I,\alpha}\left(x\right)^k$$

Using the fact that $T_{I,\alpha}$ agrees with I on constant polynomials and sends $x \mapsto \alpha$.

$$T_{(I,\alpha)}(p(x)) = \sum_k I\left(a_k\right) \cdot \alpha^k$$

$$T_{(I,\alpha)}(p(x)) = \sum_k a_k \cdot \alpha^k$$

Indeed, the right hand side is in R which is what we expect from an evaluation map.

Composition Composition in lean4 is given by

```
def comp (p q : R[X]) : R[X] :=
  p.eval2p C q
```

Notationally we can represent with

$$T_{(C,q)}(p(x)) = T_{C,q}\left(\sum_k a_k x^k\right)$$

Where C is the constant homomorphism and q is a polynomial in $R[X]$

Prop: $T_{C,q}(p)(x) = (q \circ p)(x)$

Proof: Observe,

$$T_{(C,q)}(p(x)) = T_{C,q}\left(\sum_k a_k x^k\right)$$

Using the fact that T itself is homomorphic

$$T_{(C,q)}(p(x)) = \sum_k T_{C,q}\left(a_k\right) \cdot T_{C,q}\left(x\right)^k$$

Using the fact that T acts as C on constants in R and acts maps $x \mapsto q$

$$T_{(C,q)}(p(x)) = \sum_k C\left(a_k\right) \cdot q^k$$

Finally, use the fact that for all $\alpha \in R, C(\alpha) = \alpha x^0 \in R[X]$

$$T_{(C,q)}(p(x)) = \sum_k a_k x^0 \cdot q^k$$

Indeed, $T_{(C,q)}$ is a homomorphic map that has the effect of composition of polynomials $p, q \in R[X]$. $\square$